



SmartExam

Step-by-Step Feature Testing Guide

For examiners & supervisors — how to verify every FYP-1 feature works

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Document 3 of 3 · Generated July 2026 · Each test lists steps, the expected result, and the negative check

Before You Start — One-Time Setup

Launch the three parts

1. **API server:** in `SmartExam_Root/Backend_API` run `dotnet run` . Wait until `http://localhost:5050/api/health` shows `{"status": "Healthy"}` .
2. **Web panel:** in `SmartExam_Root/Admin_Web_Panel` run `npm run dev` , then open `http://localhost:5173` .
3. **Student app:** run `Student_Desktop_App/bin/Debug/net9.0-windows/Student_Desktop_App.exe` .

Demo accounts

ROLE	EMAIL	PASSWORD
Administrator	admin@smartexam.com	Admin@123
Teacher	teacher@smartexam.com	Teacher@123
Student (Ali)	ali@smartexam.com	Student@123
Student (Sara)	sara@smartexam.com	Student@123

Tip: to start from a clean state, log in as admin and the Audit/Users pages let you reset. A full data reset is available at `POST /api/admin/reset-demo-data` (admin token). Seats are named R1C1, R1C2.

Module 1 — Authentication & Device Binding

TC-1.1 — Role-based login & JWT

ADMIN

TEACHER

STUDENT

1. Open the web panel, log in as each of admin and teacher.

Expected: Admin lands on the Admin dashboard; Teacher lands on the Teacher dashboard. Each sidebar only shows that role's pages.

Negative: enter a wrong password → "Invalid email or password", you stay on the login screen.

TC-1.2 — HWID device binding & mismatch block

STUDENT

1. In the student desktop app, log in as **ali@smartexam.com**. (First login binds this PC.)
2. Log out, log back in on the same PC.

Expected: first login binds and opens the dashboard; the second login succeeds because the hardware matches.

Negative (proxy attempt): from the web panel as admin, open **Device Bindings** — Ali now shows **Bound** with his PC name + Windows user. If a different machine tries Ali's account, login is refused with "bound to a different device", and the attempt appears in **Audit Logs** as **DEVICE_MISMATCH**.

TC-1.3 — See who is on which PC ADMIN

1. As admin, open **Device Bindings**.

Expected: each bound student row shows the real **Machine (PC name)**, the **Windows user** logged into that PC, and the **last-seen** time — so you know exactly who is sitting where.

TC-1.4 — Reset binding & force logout ADMIN

1. As admin → Device Bindings → on Ali click **Reset Binding**, confirm.
2. Then click **Force Logout** on Ali.

Expected: binding cleared (Ali re-binds on next login). If Ali's desktop app is open, within ~15 seconds it returns to the login screen with "session was ended by an administrator". Both actions appear in Audit Logs.

TC-1.5 — Change password ADMIN TEACHER

1. Click your avatar (top-right) → **Change Password**.
2. Enter current + a new password (≥ 8 chars), submit. Log out and back in with the new password.

Expected: "Password updated"; the new password works, the old one no longer does.

Negative: a wrong current password → "Current password is incorrect".

TC-1.6 — Audit trail ADMIN

1. As admin, open **Audit Logs**; filter by event type.

Expected: a live, timestamped list of logins, device decisions, and every admin action with the actor's name.

Negative: a teacher/student has no Audit Logs page and the endpoint returns 403.

Module 7 — User & Role Management

TC-7.1 — Create, edit, deactivate, delete a user ADMIN

1. Users → **Add User**; create a Student. Confirm it appears in the table.
2. Open its : menu → **Edit User**; change the name and role (try **Invigilator**); Save.
3. : → **Deactivate** → status dot goes grey. : → **Activate** → back to active.
4. : → **Delete User** on this throwaway account → it disappears.

Expected: every change persists and is reflected immediately; the Invigilator role shows an amber badge and its own tab.

Negative: editing to an email that already exists → "Email already in use" (409). Trying to **delete the seeded Teacher** → blocked ("owns course sections"). Deleting your own admin account → blocked.

TC-7.2 — Deactivation logs the student out live ADMIN STUDENT

1. Have Ali logged into the desktop app. As admin, Users → Ali : → **Deactivate**.

Expected: within seconds Ali's app kicks back to login; a fresh login is refused until reactivated.

TC-7.3 — Batch CSV import ADMIN

1. Create a text file with header `name,email,password,role` and a couple of rows.
2. Users → **Upload CSV** → pick the file.

Expected: a summary "X created, Y skipped"; new users appear. Rows with duplicate emails or missing fields are skipped with a reason.

Negative: a teacher account cannot import (endpoint 403).

Module 8 — Exam Creation & Configuration

TC-8.1 — Create an exam with all three question types TEACHER

1. Teacher → **Create Exam**. Step 1: title, **instructions**, section, date/time, duration.
2. Step 2: add a **Coding** question (+ a test case), a **Theory** question (+ optional reference answer), and an **MCQ** (≥2 options, tick the correct one).
3. Step 3: allowed apps, plagiarism threshold. Step 4: assign students. Create.

Expected: exam saved and listed on the dashboard as Scheduled. Re-open it — everything (including instructions and MCQ options) is intact.

Negative: try to proceed with no questions, an MCQ that has one option, or an MCQ with no correct answer marked → each is blocked with a clear message.

TC-8.2 — Edit & delete an exam TEACHER

1. Dashboard → **Edit** on a Scheduled exam → wizard opens pre-filled → change the title → Save Changes.
2. Dashboard → **Delete** on a Scheduled exam → confirm.

Expected: edits persist; deleted exam disappears.

Negative: editing/deleting is only allowed while Scheduled — once an exam is Active/Ended the API refuses it (409).

TC-8.3 — Answer key is never leaked to students STUDENT

1. As a student, fetch/open the assigned exam.

Expected: the MCQ shows its options but **not** which is correct; hidden test cases and reference answers are not present in the student's data.

Module 9 — Eligibility, Seating & Access Control

TC-9.1 — Register a lab & workstations ADMIN

1. Admin → **Labs & Workstations** → **New Lab** → create one.
2. Select the lab → + **Add Workstation** → machine number (e.g. R2C1) + IP → add.

Expected: the lab and its seats appear in the grid (seats are row/column named, e.g. R1C1).

Negative: a duplicate machine number in the same lab is rejected; a teacher cannot create labs (403).

TC-9.2 — Assign seats & catch conflicts TEACHER

1. Teacher → **Eligibility** → pick an exam. In each student row, choose a **Workstation** from the dropdown.
2. Try to give a second student the **same** seat.

Expected: the first assignment saves; the seat shows on the student's desktop dashboard.

Negative: the duplicate seat is rejected with "already assigned to another student".

TC-9.3 — Eligibility + add/remove students TEACHER

1. On the Eligibility page toggle a student ineligible, add a note, Save.
2. Use "**Add student...**" to add an enrolled-but-unassigned student; use the trash icon to remove one.

Expected: an ineligible student cannot start the exam; add/remove update the roster; changes are audited.

Negative: removing a student who already started a session is blocked ("mark ineligible instead").

Module 3 — Student Exam Dashboard (Desktop Client)

TC-3.1 — Assigned exam & readiness flow STUDENT

1. Log into the desktop app as a student assigned to the demo exam.
2. Confirm the exam name, instructions, permitted-app chips, Lab, Seat and Proctor are shown, and the countdown is ticking.
3. Click **Check In** (attendance) → tick turns green. Click **Verify Identity** → tick turns green.
4. Observe the **Start Exam** button.

Expected: Start Exam is disabled until **both** checks are green, then it becomes clickable. Starting the exam flips status to In Progress and begins monitoring ("Heartbeat OK").

Negative: if you are marked ineligible (TC-9.3), the dashboard says so and Start never unlocks.

TC-3.2 — Connection & low-time indicators STUDENT

1. Watch the connection pill; let the countdown run low (or set a short-duration exam).

Expected: the pill is green when connected (red if the server is unreachable); the countdown turns red under 5 minutes.

Module 10 — Live Proctoring Dashboard

TC-10.1 — Live grid shows who is on which PC

TEACHER

STUDENT

1. Start the exam on the student PC (TC-3.1). As the teacher, open **Live Monitor**.

Expected: a live card appears for the student showing their name, **seat + real PC name + Windows account**, active window, progress, and a green "Stable" status.

TC-10.2 — Focus-loss violation

TEACHER

STUDENT

1. With the exam running, on the student PC Alt-Tab away to another window for ~10 seconds.
2. Watch the teacher's Live Monitor.

Expected: the student's card turns **red** with an ALERTS badge; opening the card's timeline shows a "Focus Loss" entry. (Repeated switching is rate-limited to one alert per 30 s.)

TC-10.3 — Warning, extend time, force submit

TEACHER

STUDENT

1. On the teacher Live Monitor, click a student card → type a message → **Send warning**.
2. Click **Extend Time** → enter 15 minutes.
3. Click **Force Submit** on the student (or **Force End Exam** for all).

Expected: the warning appears in the student app's **Proctor Messages** panel (unread badge); the student's countdown jumps up by 15 minutes on its next heartbeat; force-submit ends that student's session.

Negative: extend-time with 0 or >180 minutes is rejected; a student calling these actions gets 403; a student cannot post telemetry into another student's session (blocked).

Quick Coverage Checklist

MODULE	TESTS	CORE THING TO SEE WORKING
1 — Auth & HWID	TC-1.1 ... 1.6	Login, device binding + mismatch block, who's-on-which-PC, reset/force-logout, change password, audit
7 — User & Roles	TC-7.1 ... 7.3	CRUD + guards, live deactivation, Invigilator role, CSV import
8 — Exam Creation	TC-8.1 ... 8.3	Coding/Theory/MCQ authoring, edit/delete, answer-key protection
9 — Seating & Access	TC-9.1 ... 9.3	Labs/workstations, seat assignment + conflict, eligibility, roster changes
3 — Student Dashboard	TC-3.1 ... 3.2	Assigned exam, pre-exam checks unlock Start, live indicators
10 — Live Proctoring	TC-10.1 ... 10.3	Live grid w/ PC identity, focus-loss alerts, warning/extend/force-submit